CSE 270 Sports Analytics

Predicting with distributions

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July 14, 2018

Predicting with distributions

Suppose we have two variables with normal distribution,

- Home team score
- Away team score

$H \sim N(\mu_H, \sigma_H^2)$
$A \sim N(\mu_A, \sigma_A^2)$

Then the difference between the home and away teams scores will also have normal distribution with the following parameters (assuming this two are independent)

$\mu_d = \mu_H - m_A$
$\sigma_D^2 = \sigma_H^2 + \sigma_A^2$ Thus the difference will have a normal distribution with the following parameters
$D \sim N(\mu_H - \mu_A, \sigma_H^2 + \sigma_A^2)$

Predicting with distributions

- In NBA (National Basketball Association) points scored per game by home and away teams follow normal distribution.
- Install package SportsAnalytics270 from github.
- The package contains data for Sports analytics.

```
library(devtools)
install_github("HABET/CSE270")
library(SportsAnalytics270)
```

```
library(SportsAnalytics270)
```

Predicting with distributions

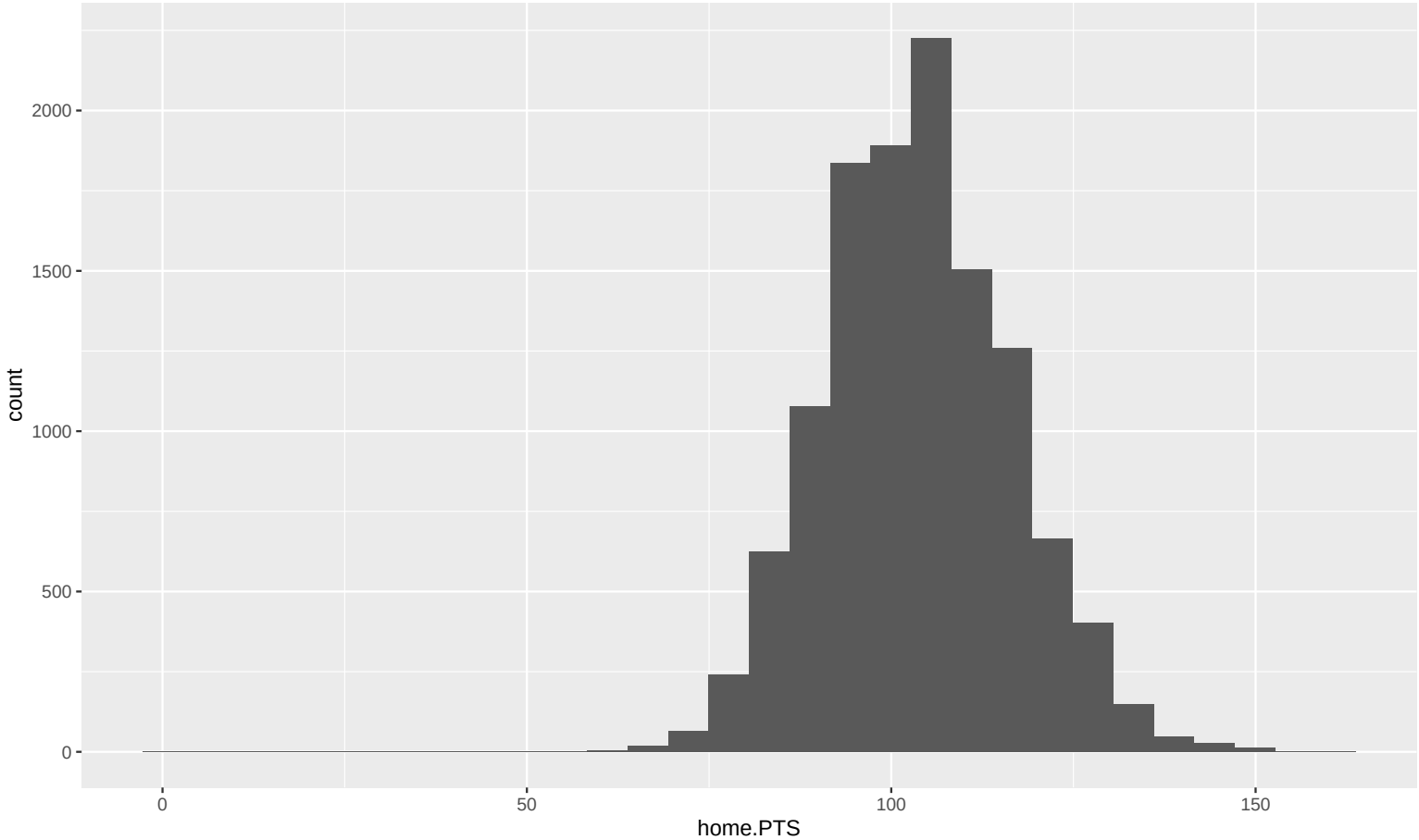
Dataframe nba2009_2018 contains NBA regular season games from 2009 to 2018- 2019 season

```
options(width = 65)
data(nba2009_2018)
head(nba2009_2018, 2)
##   SEASON_ID  GAME_DATE home.TEAM_ABBREVIATION
## 1      2009 2009-10-27                    CLE
## 2      2009 2009-10-27                    DAL
##       home.TEAM_NAME home.PTS away.TEAM_ABBREVIATION
## 1 Cleveland Cavaliers       89                    BOS
## 2    Dallas Mavericks       91                    WAS
##       away.TEAM_NAME away.PTS home.WL
## 1     Boston Celtics       95       L
## 2 Washington Wizards      102       L
```

Predicting with distributions

Plot the histogram for home points

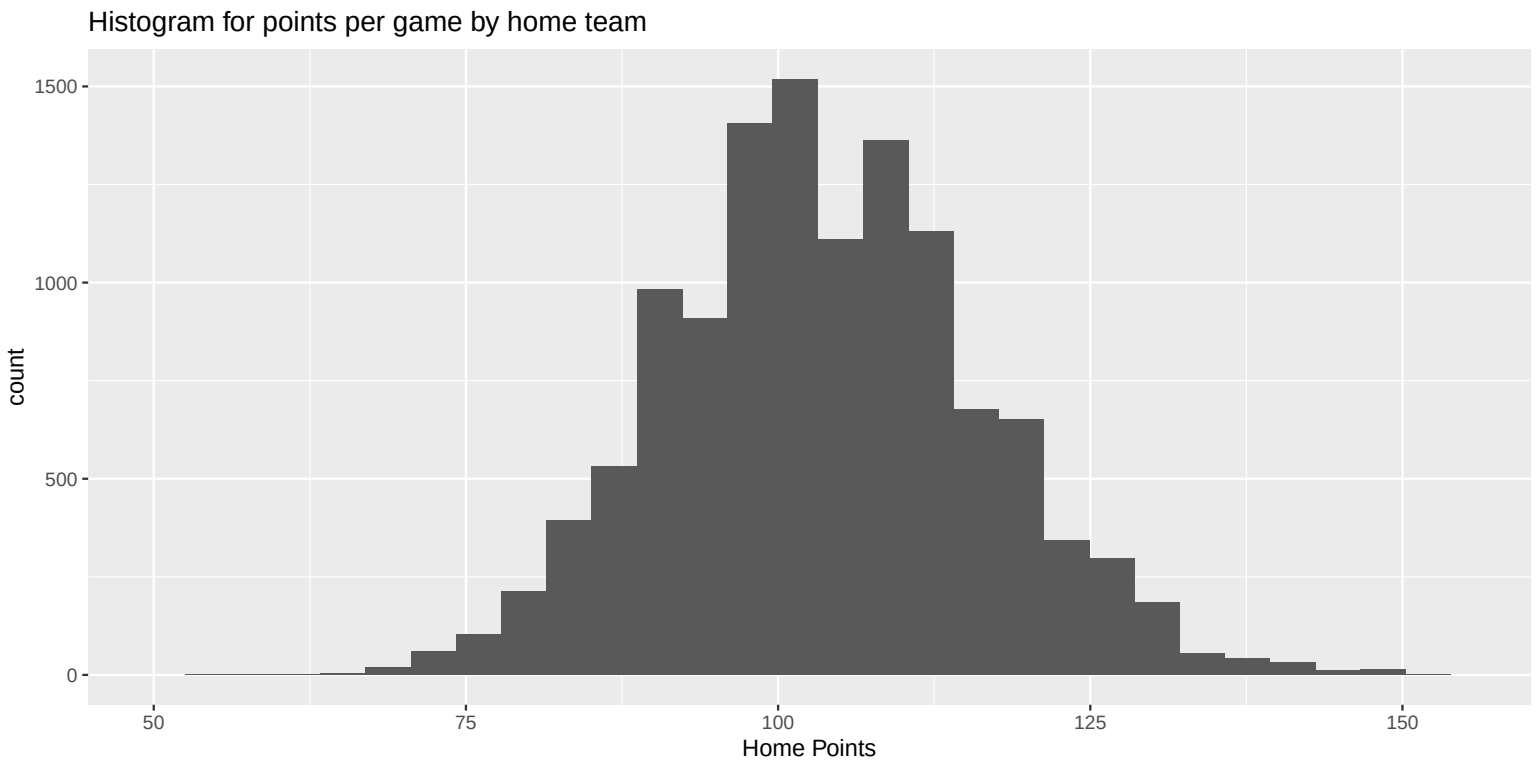
```
library(ggplot2)
ggplot(nba2009_2018, aes(x=home.PTS))+geom_histogram()
```



Predicting with distributions

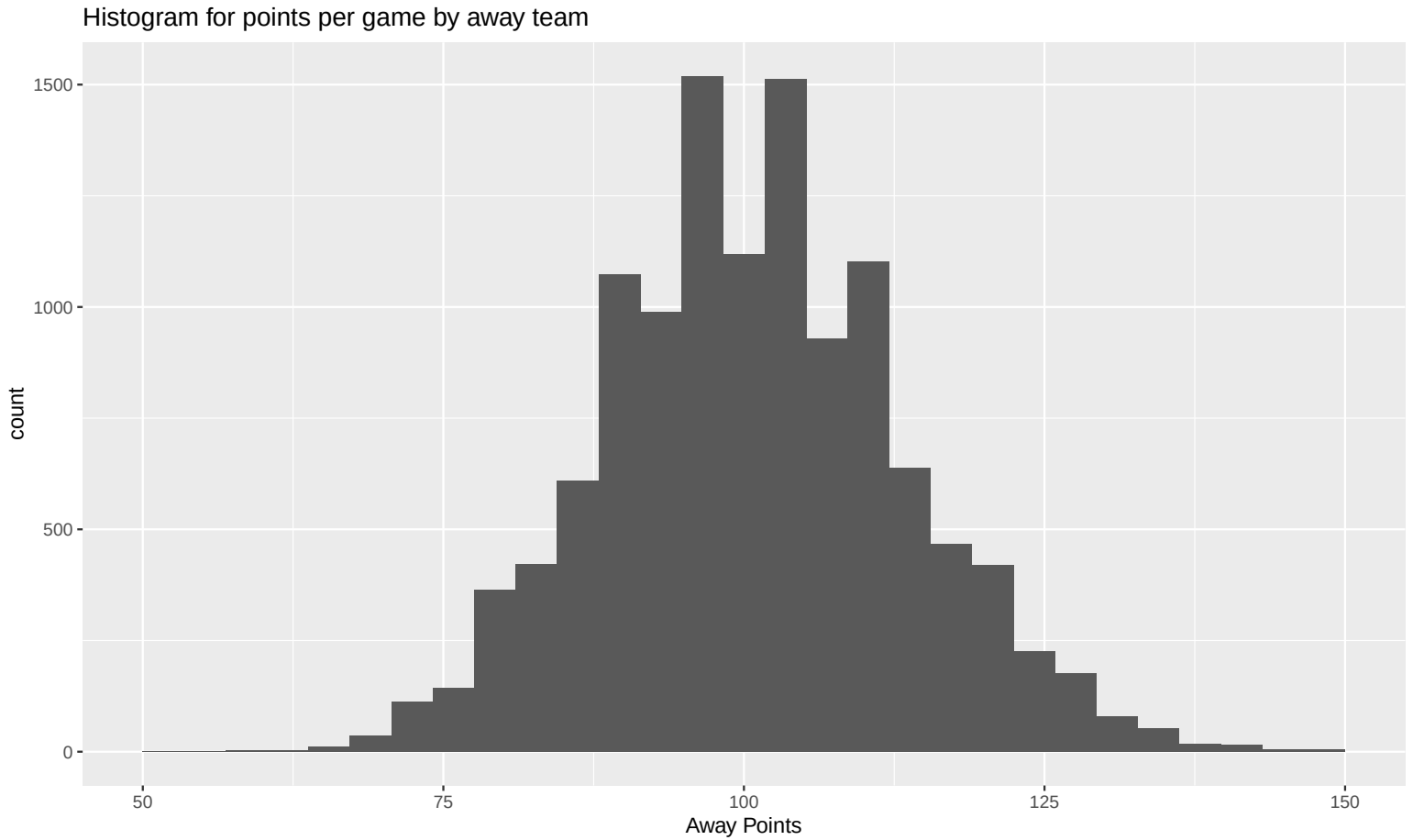
Exclude Boston-Celtics game

```
library(ggplot2)
ggplot(nba2009_2018, aes(x=home.PTS))+geom_histogram()+xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points per game by home team")
```



Predicting with distributions

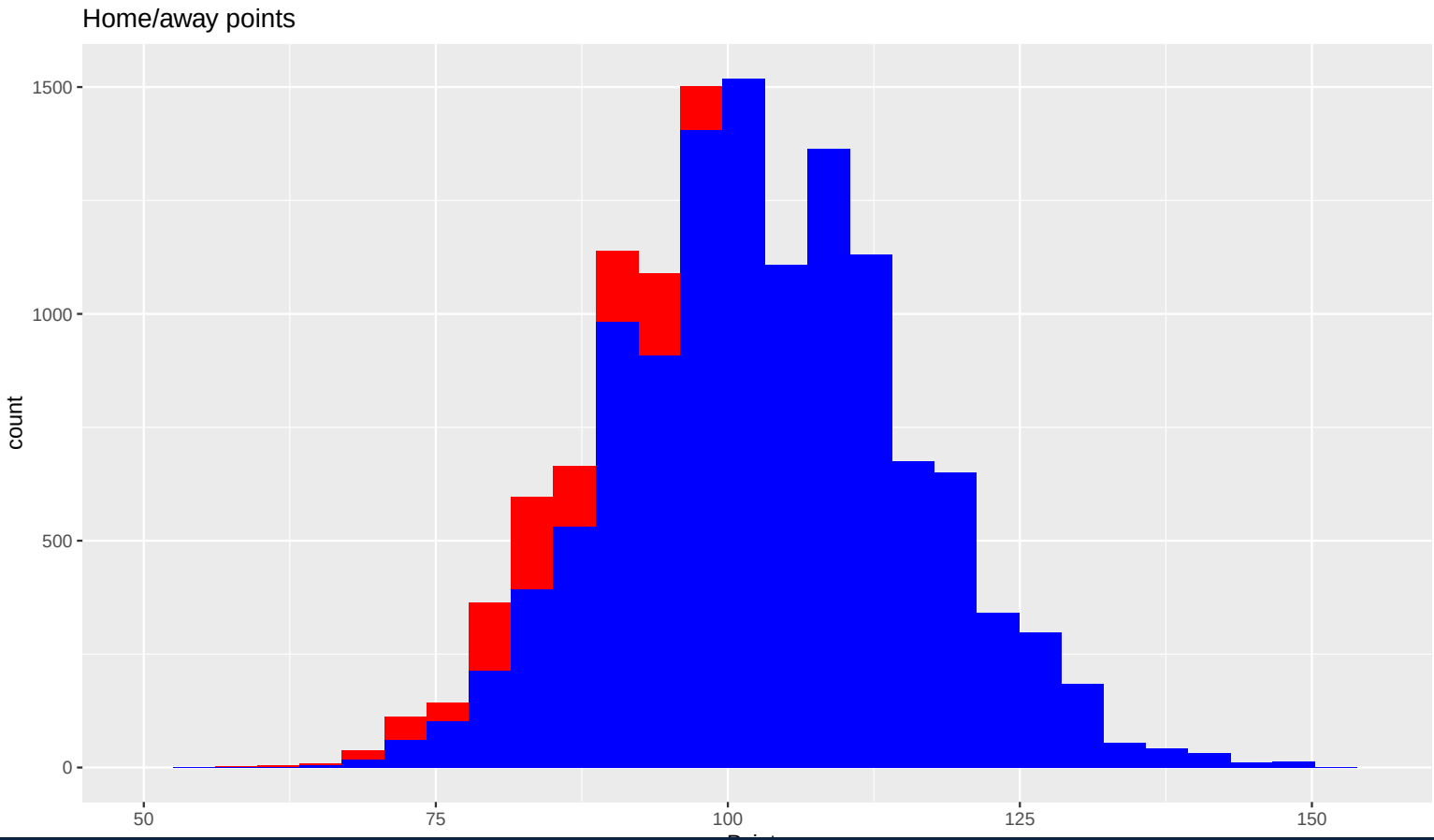
```
ggplot(nba2009_2018, aes(x=away.PTS))+geom_histogram()+xlim(c(50,150))+
  labs(x="Away Points", title="Histogram for points per game by away team")
```



Predicting with distributions

Histogram of home and away points together

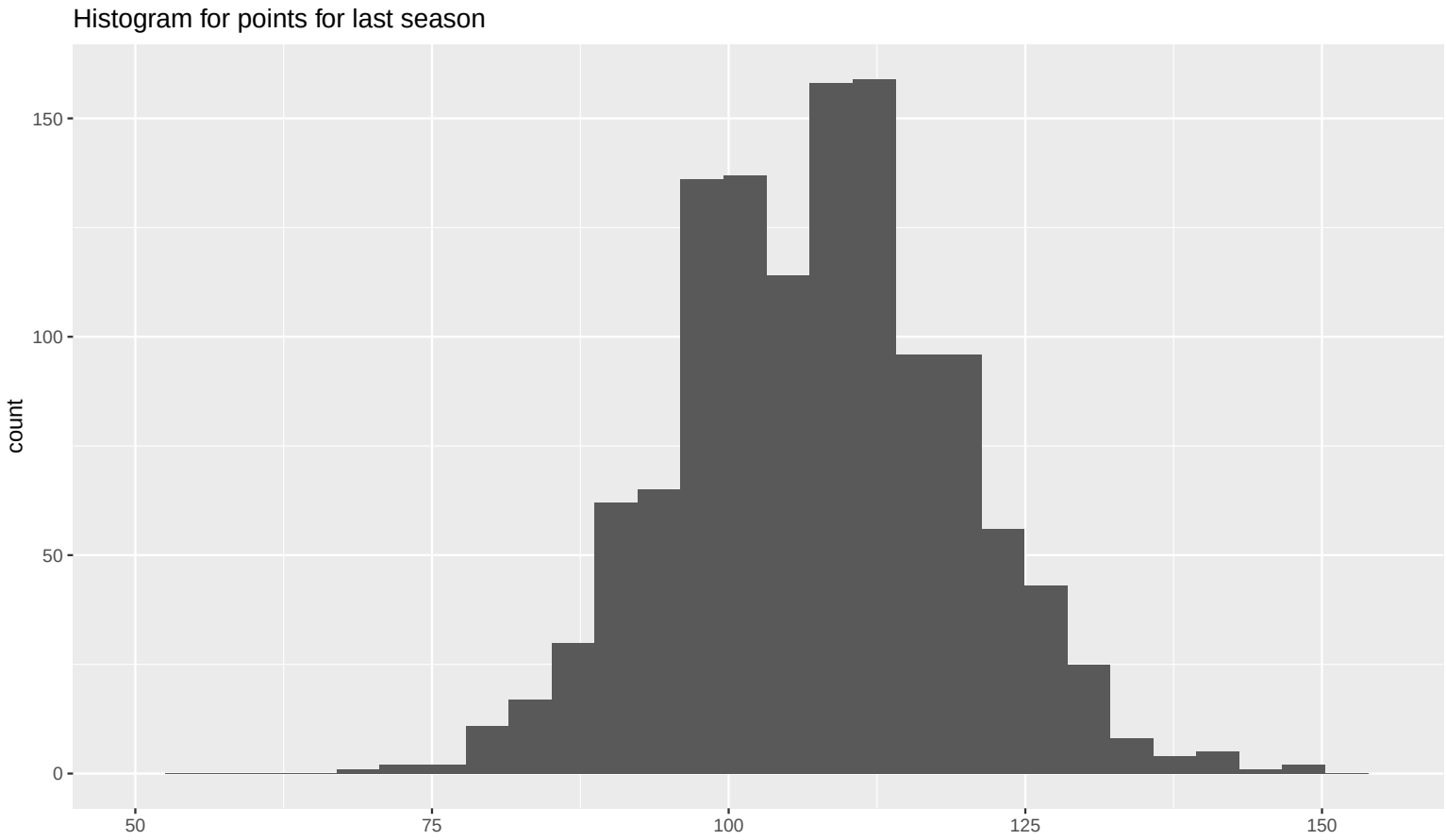
```
ggplot(nba2009_2018)+
  geom_histogram(aes(x=away.PTS), fill='red')+
  geom_histogram(aes(x=home.PTS),fill='blue')+
  xlim(c(50,155))+
  labs(x="Points", title="Home/away points")
```



Predicting with distributions

Home points for last season only

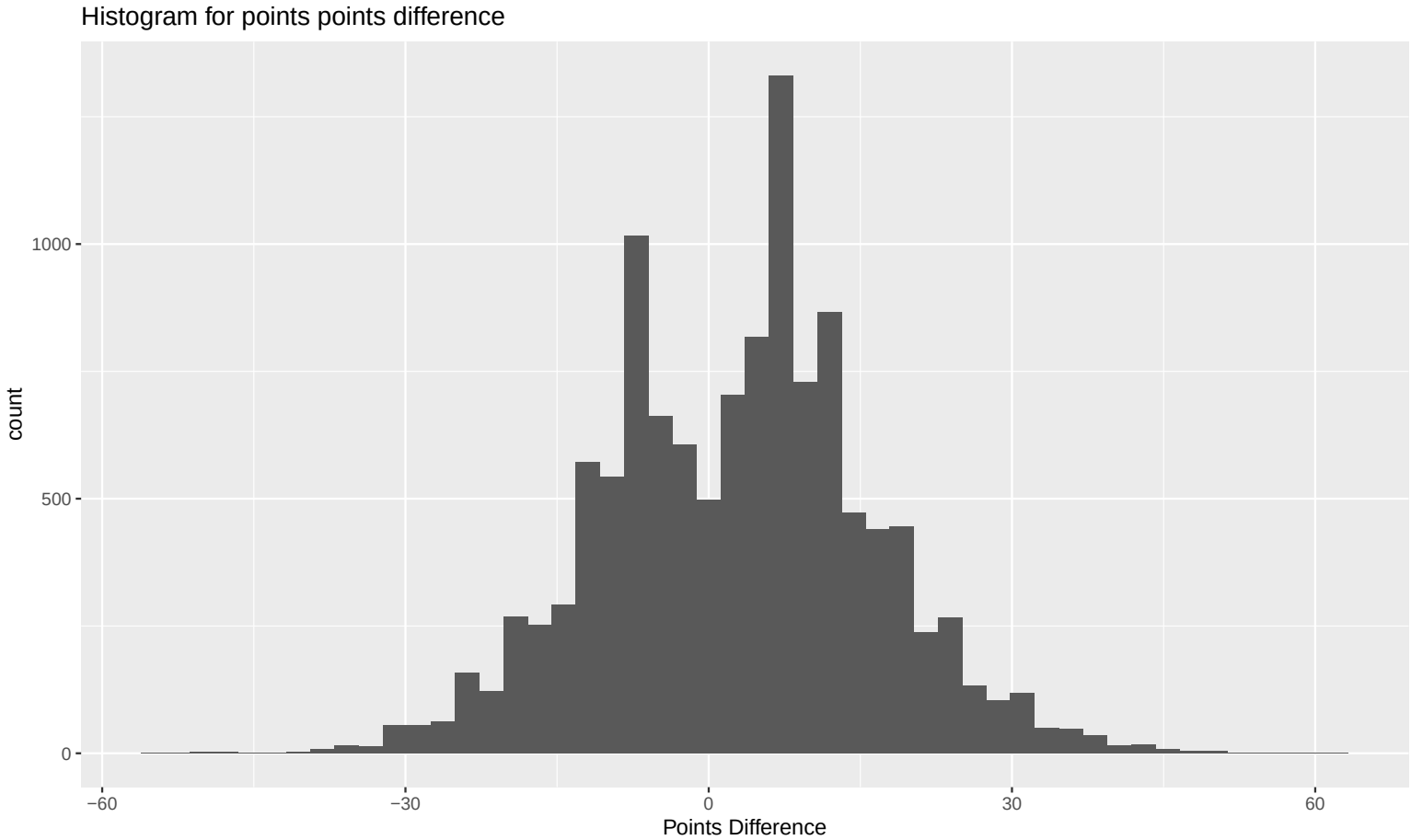
```
library(dplyr)
nba2009_2018 %>%
  filter(SEASON_ID=='2017') %>%
ggplot(aes(x=home.PTS))+geom_histogram()+xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points for last season")
```



Predicting with distributions

What about Points difference ?

```
nba2009_2018$Diff <- nba2009_2018$home.PTS-nba2009_2018$away.PTS
ggplot(nba2009_2018, aes(x=Diff))+geom_histogram(bins=50)+
  labs(x="Points Difference", title="Histogram for points points difference")
```



Predicting with distributions

Test hypothesis of Normality:

H_0 : *sample came from normal distribution*
H_1 : *sample did not come from normal distribution*
Find the p-value of the test: if is more than α 0.05 then it has a normal distribution, failed to reject H_0

Predicting with distributions

```
nba2009_2018 %>%
  filter(SEASON_ID=='2017') %>%
  group_by(home.TEAM_ABBREVIATION) %>%
  summarise(p_value = shapiro.test(home.PTS)$p.value)%>%
  arrange(p_value)
## # A tibble: 30 x 2
##    home.TEAM_ABBREVIATION p_value
##    <chr>                    <dbl>
##  1 MIA                    0.00584
##  2 NYK                    0.0629
##  3 LAL                    0.124
##  4 CHI                    0.138
##  5 MIN                    0.222
##  6 HOU                    0.250
##  7 ORL                    0.266
##  8 MIL                    0.268
##  9 NOP                    0.329
## 10 TOR                    0.329
## # ... with 20 more rows
```

Predicting with distributions

Away points: Normal distribution with $\alpha = 0.05$

```
nba2009_2018 %>%
  filter(SEASON_ID=='2017') %>%
  group_by(away.TEAM_ABBREVIATION) %>%
  summarise(p_value = shapiro.test(away.PTS)$p.value)%>%
  arrange(p_value)
## # A tibble: 30 x 2
##    away.TEAM_ABBREVIATION p_value
##    <chr>                    <dbl>
##  1 ATL                     0.0652
##  2 GSW                     0.0780
##  3 DAL                     0.126
##  4 CHA                     0.129
##  5 LAC                     0.181
##  6 MIL                     0.199
##  7 DET                     0.277
##  8 WAS                     0.329
##  9 NOP                     0.339
## 10 HOU                     0.363
## # ... with 20 more rows
```